Cambridge
O Level

Cambridge International Examinations
Cambridge Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

3948443310

MATHEMATICS (SYLLABUS D) **4024/11**

Paper 1 **October/November 2018**

2 hours

Candidates answer on the Question Paper.

Additional Materials: Geometrical instruments

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

If working is needed for any question it must be shown in the space below that question.
Omission of essential working will result in loss of marks.

ELECTRONIC CALCULATORS MUST NOT BE USED IN THIS PAPER.

The number of marks is given in brackets [ ] at the end of each question or part question.
The total of the marks for this paper is 80.

This document consists of **19** printed pages and **1** blank page.

© UCLES 2018

CAMBRIDGE International Examinations

ELECTRONIC CALCULATORS MUST NOT BE USED IN THIS PAPER

1 **(a)** Evaluate $\frac{2}{7}+\frac{1}{5}$.

Answer ..... [1]

(b) Evaluate $\frac{2}{5}\times 1\frac{1}{5}$.

Answer ..... [1]

2 **(a)** Write $17\frac{1}{2}\%$ as a fraction in its simplest form.

Answer ..... [1]

(b) Evaluate $6+4(1-0.4)$.

Answer ..... [1]

© UCLES 2018

3 y is directly proportional to the square of x.

Given that $y = 8$ when $x = 4$, find y when $x = 3$.

Answer $y =$ ..... [2]

4 During one year, the mass of a child increased from 25 kg to 30 kg.

Calculate the percentage increase in the mass.

Answer ..... % [2]

5 **(a)** Factorise completely $15a + 3ab$.

Answer ..... [1]

(b) Factorise $6k - xy + 2kx - 3y$.

Answer ..... [2]

6 $\mathrm{f}(x) = \dfrac{3}{x+4}$

(a) Find $\mathrm{f}(-6)$.

Answer ..... [1]

(b) Find $\mathrm{f}^{-1}(x)$.

Answer $\mathrm{f}^{-1}(x) =$ ..... [2]

© UCLES 2018

7 The value of each term of a sequence is 4 more than the value of the term before it.
The third term is 17 and the fourth term is 21.

(a) Find the first term.

Answer ..... [1]

(b) Find an expression for the nth term of this sequence.
Give your answer in its simplest form.

Answer ..... [2]

8 **(a)** Write down an irrational number which has a value between 4 and 5.

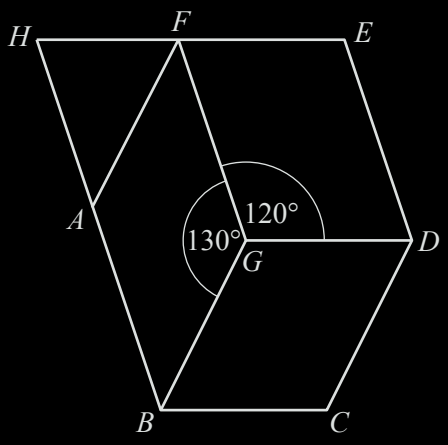
Answer ..... [1]

(b) Kofi is using number cards to form a 5-digit number.
His number is a multiple of 8.
Complete the final digit of his number.

1	2	3	4	

[1]

9



In the diagram, $AFGB$, $BGDC$ and $FEDG$ are parallelograms.
EF and BA produced meet at H.
$B\hat{G}F = 130°$ and $D\hat{G}F = 120°$.

(a) Find $B\hat{G}D$.

Answer $B\hat{G}D =$ ..... [1]

(b) Find $A\hat{B}G$.

Answer $A\hat{B}G =$ ..... [1]

(c) Find $H\hat{F}G$.

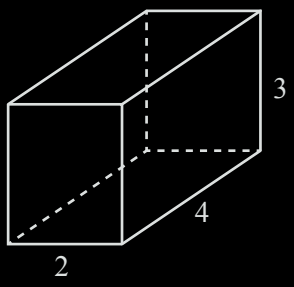
Answer $H\hat{F}G =$ ..... [1]

(d) Find $F\hat{H}A$.

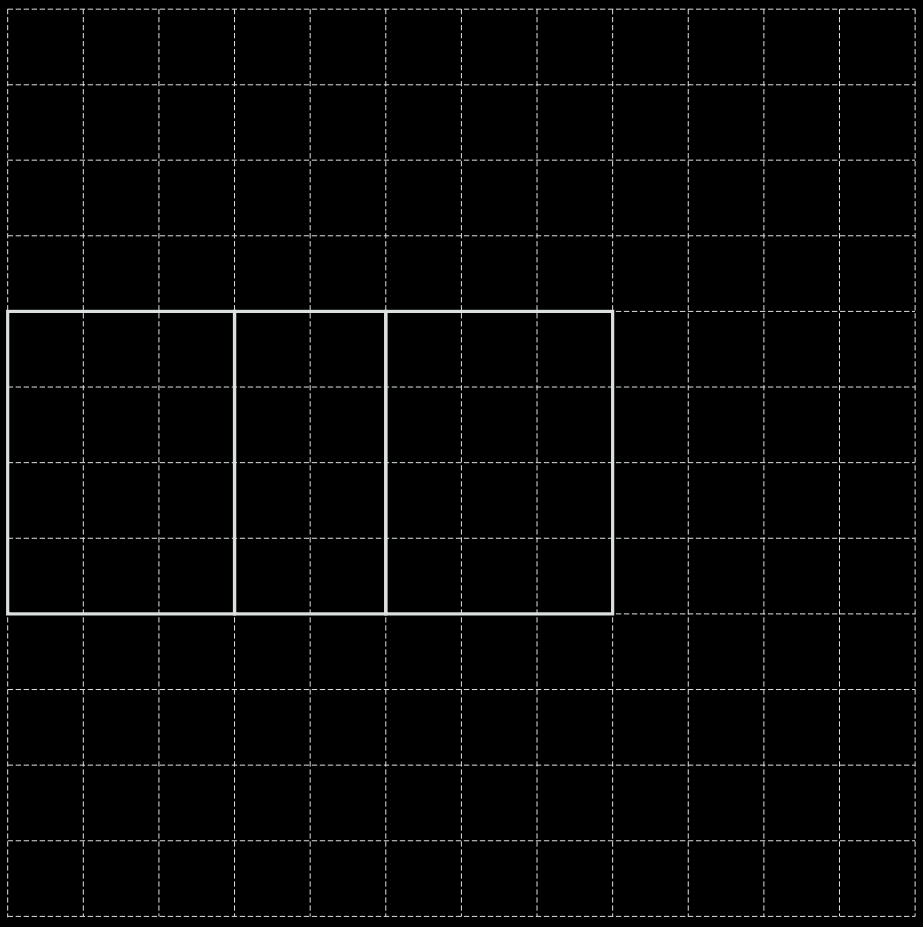
Answer $F\hat{H}A =$ ..... [1]

© UCLES 2018

10 The diagram shows a **closed** box.
The box is a cuboid.
The measurements are in centimetres.



On the grid below, complete an accurate drawing of the net of the box.
Do **not** draw outside the grid.



[2]

11 $N = 4 \times 10^5$

Giving your answers in standard form, find

(a) N^2,

Answer ..... [2]

(b) $\frac{1}{N}$.

Answer ..... [2]

12 By writing each number correct to 1 significant figure, calculate an estimate for the value of

$$\frac{614.2 \times 0.0304}{19.88}.$$

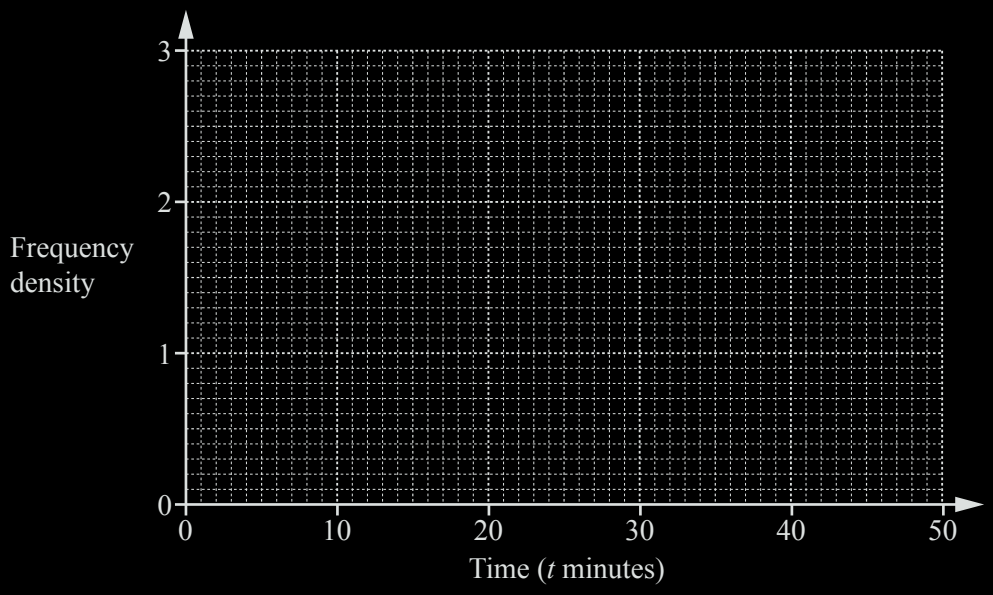
Answer ..... [2]

© UCLES 2018

13 The lengths of the times of telephone calls made by Ellie during one week are summarised in the table.

Time (t minutes)	$0 < t \leqslant 10$	$10 < t \leqslant 20$	$20 < t \leqslant 25$	$25 < t \leqslant 30$	$30 < t \leqslant 50$
Frequency	10	15	10	12	16

On the grid, draw a histogram to illustrate the distribution of these times.



[3]

14 **(a)** Evaluate $\left(\frac{1}{3}\right)^0 - \left(\frac{1}{3}\right)^2$.

Answer ..... [1]

(b) Simplify $\left(\frac{27}{x^6}\right)^{-\frac{1}{3}}$.

Answer ..... [2]

15 A five-sided spinner is numbered 1, 2, 3, 4 and 5.
Ashraf spun the spinner 200 times.
The results are shown in the table.

Number spinner lands on	1	2	3	4	5
Frequency	30	25	50	55	40

(a) Calculate the relative frequency that the spinner lands on 3.

Answer ..... [1]

(b) Meriam spins the spinner 20 times.

How many times would you expect the spinner to land on 3?

Answer ..... [1]

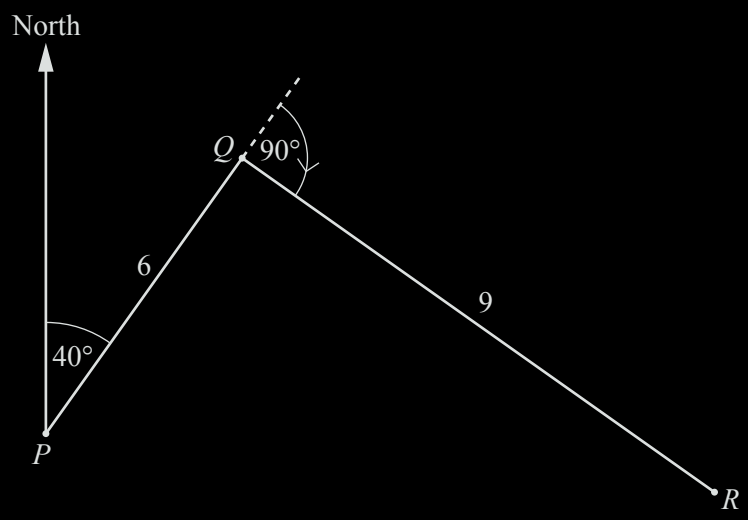
(c) Ashraf claims: "My results show that the spinner is fair".
Is his claim correct?
Give a reason for your answer.

..... because .....

..... [1]

© UCLES 2018

16



A boat travels from P to Q.
At Q, it turns through 90° and travels to R as shown in the diagram.

It then returns from R to Q, and then to P, following the same route in reverse.
$PQ = 6$ km and $QR = 9$ km.

The first part of the journey, from P to Q to R, takes 3 hours.
The return part of the journey, from R to Q to P, takes 2 hours.

(a) Calculate the average speed for the whole journey from P to Q to R and back from R to Q to P.

Answer ..... km / h [2]

(b) The bearing of Q from P is 040°.

(i) Calculate the bearing of R from Q.

Answer ..... [1]

(ii) Calculate the bearing of P from Q.

Answer ..... [1]

© UCLES 2018

17 $120 = 2^3 \times 3 \times 5$

(a) Express 1200 as the product of its prime factors.

Answer ..... [1]

(b) Find the smallest value of n, such that $120n$ is a square number.

Answer ..... [1]

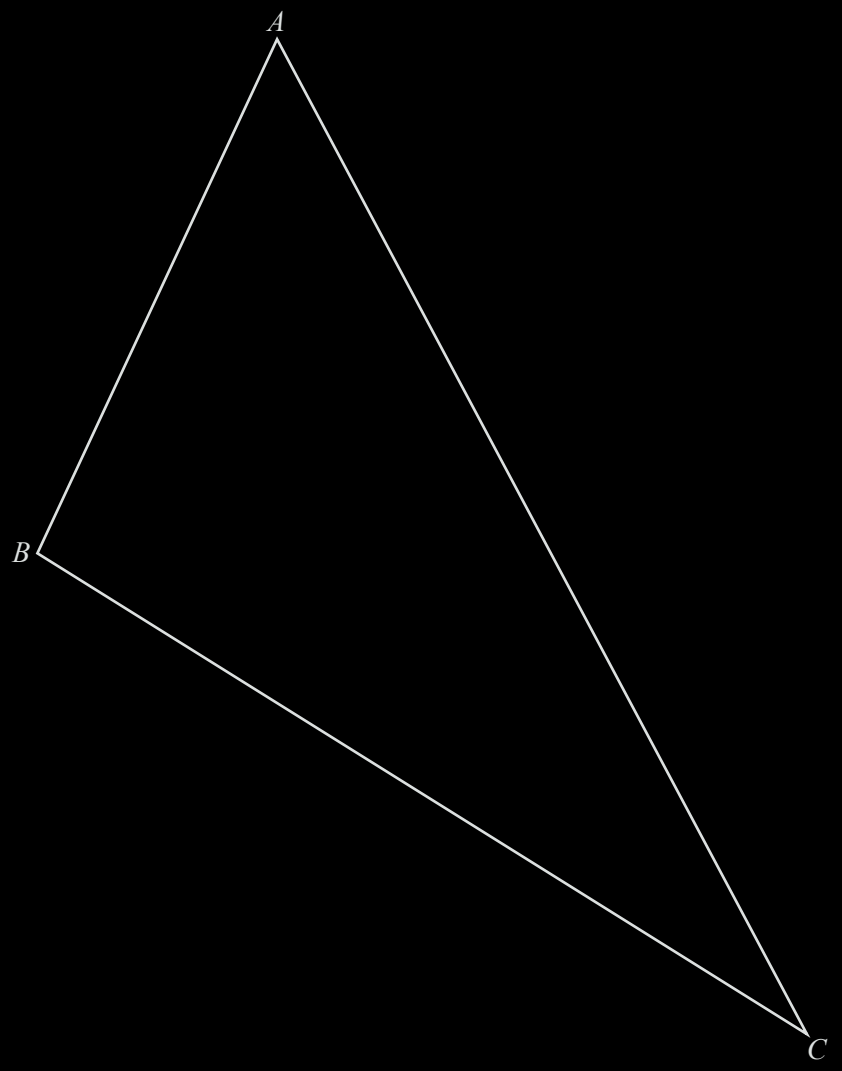
18 Four interior angles of a hexagon are 100°, 110°, 120° and 140°.
The remaining two interior angles are equal to each other.

Calculate the size of one of these interior angles.

Answer ..... [3]

© UCLES 2018

19



(a) Measure angle ABC.

Answer ..... [1]

(b) In this part, use a pair of compasses and a straight edge only.

(i) Construct the bisector of angle BAC. [2]

(ii) Construct the perpendicular bisector of AB. [2]

© UCLES 2018

20 $\mathbf{A} = \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix}$ $\mathbf{B} = \begin{pmatrix} 4 & 3 \\ 6 & -2 \end{pmatrix}$

(a) Express $2\mathbf{A} - \mathbf{B}$ as a single matrix.

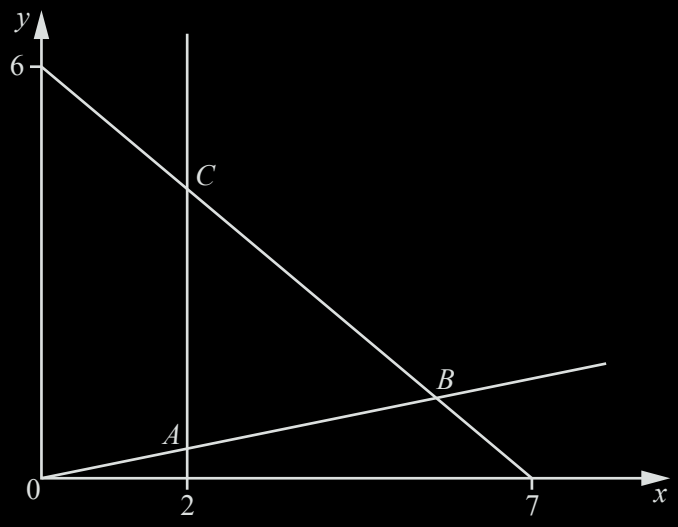
Answer $\begin{pmatrix} & \\ & \end{pmatrix}$ [2]

(b) Find $\mathbf{A}^{-1}$.

Answer $\begin{pmatrix} & \\ & \end{pmatrix}$ [2]

© UCLES 2018

21



In the diagram, the equation of the line

- through B and C is $6x + 7y = 42$
- through A and B is $y = \frac{x}{5}$.

(a) The region **inside** triangle ABC is defined by three inequalities.
One of these is $y > \frac{x}{5}$.

Write down the other two inequalities.

Answer .....

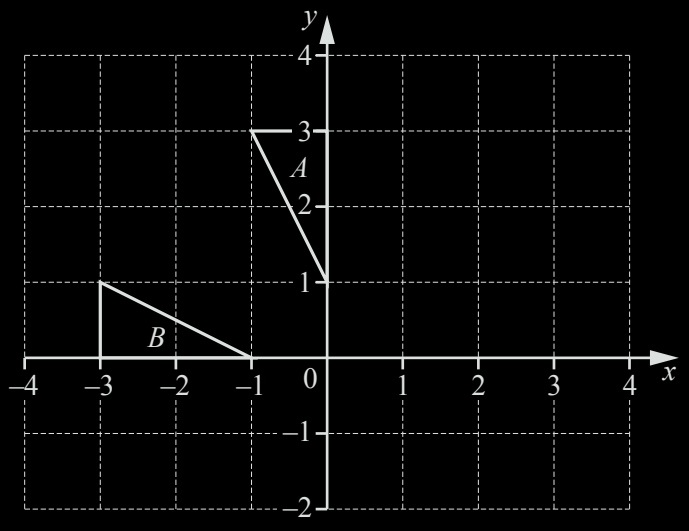
..... [2]

(b) The line $y = kx$ passes through triangle ABC.

Find all the possible **integer** values of k.

Answer ..... [2]

22



The diagram shows triangles A and B.

(a) Describe fully the **single** transformation that maps triangle A onto triangle B.

.....

..... [2]

(b) Triangle A is mapped onto triangle C by a rotation, through 90° clockwise, centre (0, 0).

Draw, and label, triangle C on the diagram. [2]

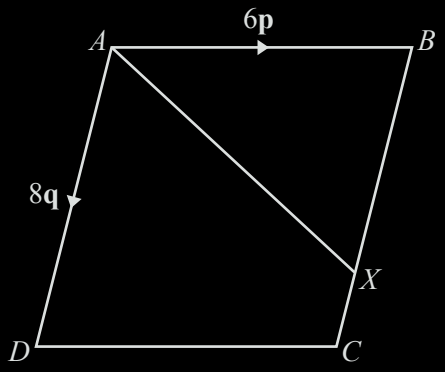
(c) Triangle B is mapped onto triangle C by the transformation T.

Find the matrix that represents the transformation T.

Answer $\begin{pmatrix} & \\ & \end{pmatrix}$ [1]

© UCLES 2018

23



In the diagram, $ABCD$ is a parallelogram.
X is the point on BC such that $BX : XC = 3 : 1$.
$\overrightarrow{AB} = 6\mathbf{p}$ and $\overrightarrow{AD} = 8\mathbf{q}$.

(a) Express $\overrightarrow{BX}$ in terms of **p** and/or **q**.

Answer ..... [1]

(b) Express $\overrightarrow{AX}$ in terms of **p** and/or **q**.

Answer ..... [1]

(c) Y is the point such that $\overrightarrow{CY} = 3\mathbf{p} + \mathbf{q}$.

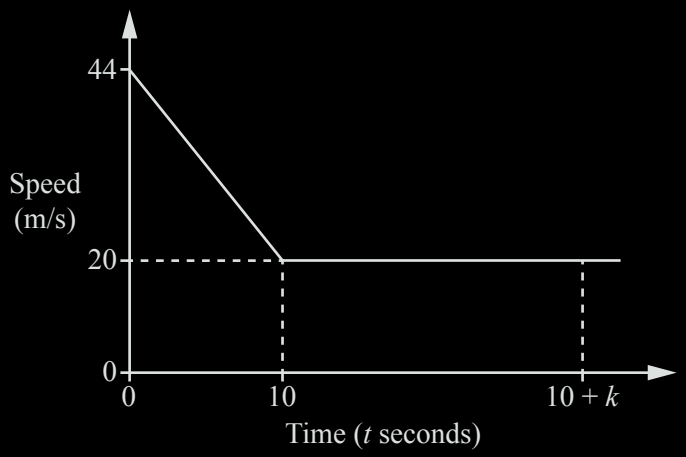
(i) Express $\overrightarrow{AY}$ in terms of **p** and/or **q**.

Answer ..... [1]

(ii) Find the ratio $AX : AY$.

Answer ..... : ..... [1]

24 The diagram is the speed–time graph of part of a train's journey.



The train slows down uniformly from a speed of 44 m/s to a speed of 20 m/s in a time of 10 seconds. It then continues at a constant speed of 20 m/s.

(a) Find the deceleration when $t = 5$.

Answer ..... m/s^2 [1]

(b) Find the speed when $t = 5$.

Answer ..... m/s [1]

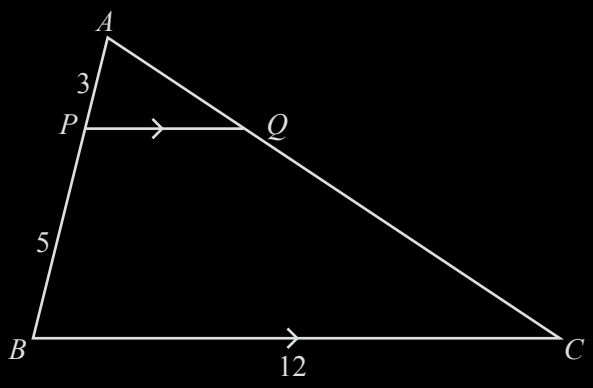
(c) The distance travelled from $t = 0$ to $t = 10$ is equal to the distance travelled from $t = 10$ to $t = 10 + k$.

Find k.

Answer $k =$ ..... [3]

© UCLES 2018

25



In the triangle ABC, P and Q are points on AB and AC such that PQ is parallel to BC.

$AP = 3$ cm, $PB = 5$ cm and $BC = 12$ cm.

(a) Find PQ.

Answer ..... cm [2]

(b) The area of triangle ABC is x cm^2.

Find an expression, in terms of x, for the area of trapezium $BCQP$.

Answer ..... cm^2 [2]

© UCLES 2018

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge International Examinations Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cie.org.uk after the live examination series.

Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.

© UCLES 2018